LUX

Language Uncertainty Extractor

FINAL ACADEMIC REVIEW REPORT

Pre-generation estimation of empirical answer unreliability risk in a frozen language model

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Project focus. LUX studies whether prompt information and the target model's pre-generation hidden representations can help estimate empirical answer unreliability. It combines a reproducible generation-and-evaluation cohort, comparative regression experiments, and a traceable baseline inference interface.

Abstract

LUX investigates whether information available before answer generation can estimate how often a frozen language model produces answers judged unreliable against a supplied passage. The research compares observable prompt features with prompt-conditioned hidden representations from Qwen3-1.7B, evaluating prompt-only, hidden-only, and fused predictors against constant baselines.

The cohort contains 500 SQuAD v2.0 prompts, ten generated answers per prompt, and 5,000 evaluation records. Applying a strict eligibility rule yields 486 prompts with ten successful binary judgments each. Their empirical risk is the proportion of answers labelled UNRELIABLE; the cohort mean is 0.208642 and the median is zero.

Across the expanded out-of-fold benchmark, the median baseline achieved the lowest mean absolute error. Prompt-only Support Vector Regression was the closest learned configuration. Several hidden-state and fused models showed modest gains on RMSE or R^2 relative to the mean baseline, demonstrating that model comparisons depend on the metric being considered.

1. Problem Statement and Objectives

Language models can produce fluent answers that are not supported by the evidence supplied in a prompt. LUX examines a pre-generation approach: estimate the empirical unreliability associated with a question and passage before sampling an answer.

The central research question is whether combining prompt-level features with the target model's internal pre-generation representations improves empirical unreliability-risk prediction compared with either information source alone and with constant baselines.

The project objectives are to establish a reproducible passage-grounded cohort, derive risk targets from repeated generations and evaluator judgments, compare prompt-only, hidden-only and fused feature groups, evaluate regression models using MAE, RMSE and R^2, and provide a traceable inference interface with validation and conservative routing.

2. Dataset and Experimental Methodology

2.1 Prompt source and target model

SQuAD v2.0 supplies questions paired with passages, including questions that cannot be answered from the provided context. The frozen target is Qwen/Qwen3-1.7B, pinned to revision 70d244cc86ccca08cf5af4e1e306ecf908b1ad5e. The model is used to extract prompt-conditioned representations and generate answers under a fixed protocol.

SETTING	RECORDED VALUE
Target model	Qwen/Qwen3-1.7B
Pinned revision	70d244cc86ccca08cf5af4e1e306ecf908b1ad5e
Execution device	CPU
Thinking mode	Disabled
Sampling	Enabled
Temperature / top-p / top-k	0.7 / 0.8 / 20
Maximum new tokens	512
Generations per prompt	10
Generation seeds	1001 through 1010

2.2 Evaluator labels and target calculation

A versioned Gemini evaluator assessed each generated answer against its question and supplied passage. The three labels were RELIABLE, UNRELIABLE, and INADEQUATE. For strict target construction, a prompt qualifies when all ten evaluations are successful binary judgments, with each label either RELIABLE or UNRELIABLE.

Empirical risk = number of UNRELIABLE labels ÷ 10.

The completed cohort contains 5,000 successful generations and 5,000 evaluation records. Fourteen prompts included inadequate judgments and were excluded from strict target construction, leaving 486 prompts and 4,860 binary judgments.

2.3 Cohort summary

QUANTITY	RESULT
Selected prompts	500
Generated answers	5,000
Evaluation records	5,000
Eligible prompts	486
Reliable judgments in eligible cohort	3,846
Unreliable judgments in eligible cohort	1,014
Zero-risk prompts	367 (75.51%)
Positive-risk prompts	119 (24.49%)
Mean empirical risk	0.208642
Median empirical risk	0
Observed risk range	0 to 1

3. System Architecture

LUX is organized around an offline research pipeline and a separate inference path. The research pipeline builds the cohort, extracts representations, constructs empirical risk targets and evaluates regression models. The inference interface validates an incoming prompt and target configuration, loads a versioned baseline artifact, returns a risk estimate with provenance, and applies a conservative routing policy.

OFFLINE RESEARCH PIPELINE	INFERENCE PATH
1. Validate canonical question and context	1. Validate supplied prompt
2. Lock target model and generation settings	2. Validate target identity and configuration
3. Extract five prompt-level features	3. Load and validate versioned baseline artifact
4. Run Qwen forward pass and pool hidden states	4. Return risk estimate
5. Generate ten answers and obtain evaluator labels	5. Attach provenance and warnings
6. Aggregate strict empirical risk targets	6. Apply allow, verify or escalate routing
7. Fit and evaluate regression models	Output: risk estimate and routing action

3.1 Prompt features

FEATURE	DESCRIPTION
question_chars	Character count of the question.
context_chars	Character count of the passage.
question_tokens	Question token count under the frozen tokenizer convention.
context_tokens	Context token count under the frozen tokenizer convention.
question_ends_with_question_mark	Binary indicator of whether the question ends with a question mark.

3.2 Hidden-state representations

The frozen Qwen model processes the rendered prompt before answer generation. The saved representation contains 29 returned layer entries, each represented by a 2,048-dimensional vector pooled over prompt tokens. The cohort matrix has shape (486, 29, 2048).

Experiments use two representation conventions: the final layer's token-mean vector, and a layer-axis average that produces one 2,048-dimensional vector per prompt. The expanded benchmark uses layer-axis averaging; token pooling is performed during representation capture.

3.3 Preprocessing and feature groups

The experiments compare prompt-only, hidden-only and fused features. The corrected cross-validation pipeline fits StandardScaler and PCA within each training fold, using 50 PCA components for hidden features where specified. Fusion combines the five prompt features with transformed hidden components.

4. Models and Evaluation

The expanded benchmark contains 39 saved configurations: six repeated baseline entries and 33 learned configurations across 11 learned model families and three feature groups.

MODEL FAMILY	MODEL NAMES
Constant baselines	Dummy Mean, Dummy Median
Linear regression	Ridge, Lasso, Elastic Net
Kernel regression	Support Vector Regression (SVR)
Instance-based	K-Nearest Neighbors (KNN)
Tree	Decision Tree (DT)
Tree ensembles	Random Forest (RF), Extra Trees (ET)
Boosting	Gradient Boosting Regressor (GBR), HistGradientBoosting (HGB)
Neural	Multilayer Perceptron (MLP)

Evaluation metrics. MAE measures average absolute prediction error. RMSE gives larger errors more influence. R^2 compares squared prediction error with a constant mean reference; a negative value indicates worse squared-error performance than that reference.

5. Results

5.1 Original fixed-split comparison

CONFIGURATION	TEST MAE	TEST RMSE	TEST R^2
Mean baseline	0.314021	0.383036	-0.001978
Ridge prompt	0.315697	0.384700	-0.010700
Ridge hidden	0.315337	0.394089	-0.060638

Ridge fusion	0.313035	0.393379	-0.056817
RF prompt	0.297240	0.382846	-0.000983
RF hidden	0.274877	0.401278	-0.099688
RF fusion	0.272781	0.398045	-0.082036
ET prompt	0.303767	0.390348	-0.040598
ET hidden	0.281692	0.391414	-0.046290
ET fusion	0.286089	0.394435	-0.062502
HGB prompt	0.317597	0.414066	-0.170893
HGB hidden	0.297143	0.414245	-0.171904
HGB fusion	0.299358	0.406937	-0.130922

RF fusion recorded the lowest MAE in this fixed-split comparison at 0.272781. Its R^2 was -0.082036, illustrating that a lower absolute error did not translate into better squared-error performance than the evaluated mean reference.

5.2 Corrected training-only cross-validation

MEASURE	MEAN BASELINE	SELECTED RF HIDDEN
Test MAE	0.314021	0.274877
Test RMSE	0.383036	0.401278
Test R^2	-0.001978	-0.099688

Random Forest with hidden features was selected using five-fold cross-validation within the 340-prompt training partition and evaluated on the 146-prompt test partition. Its MAE was 12.47% lower than the mean baseline on that split. RMSE and R^2 were less favorable than the baseline.

The paired bootstrap interval for the baseline-minus-model MAE difference was [0.016853, 0.059941] on this fixed test set. This result is specific to the split and is considered separately from the expanded benchmark.

5.3 Expanded out-of-fold benchmark

CONFIGURATION	FEATURES	OOF MAE	OOF RMSE	OOF R^2
Dummy Mean	Prompt	0.317064	0.386904	-0.000659
Dummy Median	Prompt	0.208642	0.439463	-0.290993
SVR	Prompt	0.214182	0.434953	-0.264634
SVR	Hidden	0.223997	0.422184	-0.191469
SVR	Fusion	0.224017	0.422202	-0.191574
Random Forest	Hidden	0.303474	0.383895	0.014843
Extra Trees	Hidden	0.310578	0.384885	0.009756
Gradient Boosting	Hidden	0.300624	0.392408	-0.029333
Extra Trees	Fusion	0.309409	0.383472	0.017014
Decision Tree	Fusion	0.296684	0.464977	-0.445247

The Dummy Median baseline achieved the lowest out-of-fold MAE at 0.208642. Prompt-only SVR was the closest learned configuration, with MAE 0.214182. The median's performance is consistent with the target distribution: 75.51% of eligible prompts have zero empirical risk.

Some learned configurations performed better on RMSE and R^2 than the mean baseline. Hidden Random Forest reached RMSE 0.383895 and R^2 0.014843, while fused Extra Trees reached RMSE 0.383472 and R^2 0.017014. Their MAEs remained above the median baseline.

5.4 Retrospective routing analysis

POLICY AT THRESHOLD 0.2	NORMAL PROMPTS	ESCALATION	NORMAL-PATH MEAN RISK	UNRELIABLE MASS CAPTURED
No estimator	486	0.00%	0.208642	0.00%
Median baseline	486	0.00%	0.208642	0.00%
Mean baseline	0	100.00%	Undefined	100.00%
Prompt SVR	486	0.00%	0.208642	0.00%
Hidden RF	287	40.95%	0.172474	51.18%
Fused Extra Trees	263	45.88%	0.169582	56.02%

The routing analysis evaluated 176 fixed-threshold policies across saved risk scores. At the illustrated threshold of 0.2, hidden Random Forest and fused Extra Trees escalated a subset of prompts and reduced the mean empirical risk among prompts left on the normal path. These figures provide a descriptive sensitivity comparison.

6. Software Implementation and Validation

6.1 Supported inference interface

The Python inference function in src/predictor/inference.py accepts a prompt, target model identity, revision, generation configuration and a loaded baseline artifact. It returns a structured prediction with provenance metadata, a baseline statistic, calibration status and warnings.

The versioned artifacts baseline_mean_v2.json and baseline_median_v2.json are tied to the frozen target specification and source fingerprints. The mean artifact returns approximately 0.208642 for each valid prompt; the median artifact returns zero. These cohort-level baselines provide reference estimates for comparison.

6.2 Validation and routing

The interface validates prompt structure, allowed fields, target identity, revision, generation settings and artifact integrity. Invalid inputs or incompatible artifacts produce explicit errors. The policy interface exposes allow, verify and escalate actions, with conservative escalation as its default behavior. Explicit thresholds are checked for valid ranges and ordering.

6.3 Recorded engineering validation results

VALIDATION MEASURE	RECORDED RESULT
End-to-end focused test suite	75 passed
Broader test suite	409 passed
Routing and related tests	91 passed
Local cold-start samples	20
Warm workload	12,000 measured requests
Cold-process median	4.267 seconds
Warm request mean	0.090 to 0.110 milliseconds
Peak working set	Approximately 116.9 MiB

The test counts come from overlapping suites in the engineering validation reports. Timing measurements describe the constant baseline interface.

7. Contributions

LUX brings together a reproducible cohort of 500 prompts, ten generations per prompt, evaluator-backed empirical targets for 486 eligible prompts, and comparative experiments across prompt-only, hidden-only and fused feature groups. The model study covers linear methods, kernel regression, instance-based learning, tree models, ensembles, boosting and a neural regressor, alongside mean and median baselines.

The project also includes saved out-of-fold benchmark results, retrospective routing sensitivity analyses, and a tested baseline inference interface with versioned artifacts, provenance metadata, input validation and conservative action selection.

8. Conclusion

LUX provides an empirical study of pre-generation answer unreliability-risk estimation for a frozen Qwen3-1.7B configuration. From 500 SQuAD v2.0 prompts and 5,000 generated answers, the strict evaluation process produced 486 empirical risk targets. Prompt-only, hidden-only and fused predictors were compared across several regression families.

In the expanded out-of-fold benchmark, the median baseline achieved the lowest MAE at 0.208642. Prompt-only SVR was the closest learned model at 0.214182. Some hidden-state and fused configurations improved RMSE or R^2 relative to the mean baseline, while their MAE remained higher than the median baseline.

The project's central contribution is an inspectable experimental pipeline supported by comparative results and a traceable baseline inference interface. The findings highlight the value of strong baselines and multiple evaluation metrics when assessing pre-generation risk prediction.

Key Project Artifacts

run_production.py; src/evaluation/prompt_cp4_7.py; src/predictor/inference.py; docs/engineering/baseline_mean_v2.json; docs/engineering/baseline_median_v2.json; experiments/final_cohort_2026_10_24/; experiments/expanded_model_selection_2026_10_24/; experiments/routing_policy_evaluation_2026_10_27/